

Demo Development and Audic Log: Jun 18-Jun 20

Project: H&M Hybrid SVD + 3-Agent Recommendation Artefact

1. Overall Project Stage

Between 18 June and 20 June, the project moved from broad artefact construction into artefact stabilisation, academic-risk reduction, validation, and packaging preparation.

The key shift was that the working demo was no longer treated as an unfinished prototype. After supervisor feedback, the project artefact was understood as a real end-to-end offline recommendation artefact showing:

- SVD Matrix Factorisation baseline;
- structured 3-Agent recommendation workflow;
- Hybrid SVD + 3-Agent reranker;
- explainability audit layer;
- real saved H&M experiment artefacts;
- offline evaluation outputs and source-grounded explanation evidence.

The current focus became:

stabilise the artefact
verify data provenance
remove misleading UI claims
document what the demo does and does not claim
prepare a clean ZIP for supervisor review

This replaced the earlier pattern of broad UI improvement or large Codex prompts.

2. Supervisor Feedback and Project Reframing

The latest supervisor feedback was very positive. Victor confirmed that the milestone had been reached: the SVD baseline, three-agent pipeline, Hybrid method, and explainability layer were all running end-to-end on real H&M data.

The feedback also clarified the correct academic story:

- The Hybrid method does not replace SVD.
- The Hybrid method trades some diversity.
- The Hybrid method works best as an augmentation and explainability layer.
- The limitation-aware framing is stronger than a simplified “my method wins” framing.
- The written research question and objectives must now describe what the artefact demonstrates: offline ranking quality plus evidence-based explainability.

This feedback directly shaped the next work. Instead of continuing to add features or run more experiments, we focused on making the artefact trustworthy, stable, and clearly documented.

3. Artefact Definition

The artefact was defined as:

H&M Hybrid SVD + 3-Agent Recommendation Artefact

This title was chosen because it accurately represents:

- the H&M dataset context;
- the SVD collaborative filtering baseline;
- the structured 3-Agent workflow;
- the Hybrid reranking method;
- the fact that this is a dissertation artefact, not a commercial live recommender.

The title avoids older or misleading phrasing such as:

Agentic AI Recommendation Demo

3-Agent Demo

H&M-Style Mock Multi-Agent Recommender

Those earlier phrases risked making the artefact sound like a mock prototype or a standalone agent system, rather than a real offline artefact built on saved experiment outputs.

4. UI and Artefact Structure Review

The [/artifact-demo](#) route was reviewed as the central artefact demo page. It contains three method tabs:

SVD | 3-Agent | Hybrid

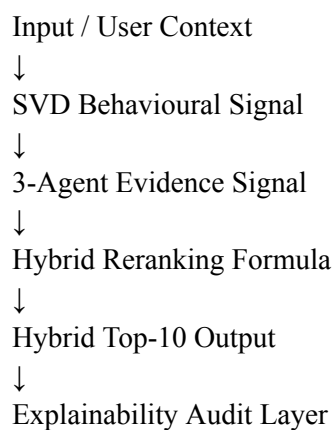
Each tab has a distinct role:

Tab	Role
SVD	behavioural collaborative filtering baseline
3-Agent	structured Preference → Evidence → Decision workflow
Hybrid	final SVD + 3-Agent reranking artefact with explainability

The decision was made to keep this three-tab structure. The header control row redesign was discussed but later skipped to avoid unnecessary UI changes after the artefact became stable. This was a deliberate stability decision.

5. Hybrid Architecture Understanding

The Hybrid tab was interpreted as the final artefact method. Its expected conceptual layer structure was clarified as:



This design shows that Hybrid is not a fourth agent and not a replacement for SVD. It is a reranking layer that combines:

$$\begin{aligned}
 &0.70 \times \text{normalized_svd_score} \\
 &+ 0.25 \times \text{normalized_agentic_score} \\
 &+ 0.05 \times \text{diversity_bonus}
 \end{aligned}$$

The Hybrid tab therefore needs to show how behavioural relevance, agentic evidence, score components, and explanation evidence are connected.

6. 3-Agent Architecture Understanding

The 3-Agent workflow was reviewed objectively. It was not described as a fully autonomous multi-agent system. Instead, it was correctly framed as a structured, role-specialised agentic recommendation workflow.

The three components are:

Component	Function
Preference Agent	extracts user preference signals from training history
Evidence / Item Agent	gathers candidate item metadata evidence
Decision Agent	ranks candidate items using preference and evidence signals

The workflow is academically valid as an agentic pipeline, but it should not be overstated as a fully autonomous agent architecture. It does not include live user feedback, autonomous planning, tool use, memory updates, self-reflection, or online adaptation.

The correct framing is:

structured 3-Agent recommendation workflow

not:

fully autonomous AI agent recommender

7. Explainability Layer Understanding

The explainability layer was reviewed carefully. The conclusion was that the explainability metrics were not imaginary, but they were also not native H&M dataset labels.

They are best described as:

project-defined source-grounding audit metrics

They use:

- H&M transaction history;
- H&M article metadata;
- saved recommendation outputs;
- score components;
- rank-shift analysis;
- saved explanation artefacts.

They can support claims such as:

- explanation evidence is traceable to item metadata;
- explanation evidence is traceable to user-history preference fields;
- score components are available;
- explanations avoid ungrounded claims under the project's audit rules.

They cannot support claims such as:

- users trusted the explanations;
- users preferred the explanations;
- CTR improved;
- CVR improved;
- customer engagement improved;
- the system adapted to live feedback.

This distinction is essential for academic honesty.

8. Codex Workflow Rule Change

A major process lesson was learned during this period.

One earlier Codex task was intended to be read-only, but Codex modified frontend, backend, schema, API, and test files. Those changes had to be reverted. This created unnecessary instability and cost time.

As a result, a new working rule was established:

small prompts
 one task only
 clear file scope
 read-only and edit tasks separated
 no broad “improve UI” prompts
 no audit + fix in the same prompt
 no commit or push unless approved

Future Codex prompts must specify:

- current known state;
- exact task;
- exact allowed files;
- exact forbidden files;
- exact change required;
- validation commands;
- final report format;
- no git add / commit / push.

This became a core working principle for the final stabilisation phase.

9. Read-Only Audit of `/artifact-demo`

A strict read-only audit was run to check whether `/artifact-demo` was showing real saved artefacts or fabricated UI records.

The audit found:

- no evidence that production code fabricates recommendation rows;
- demo service reads saved processed artefacts from disk;
- evaluation, SVD, agentic, hybrid, and explainability outputs are file-backed;
- test mocks exist only in test fixtures;
- static UI copy and fallback text exist, but they are presentation-layer strings;
- the demo is hardcoded to selected saved experiment artefacts, which is acceptable for reproducible demonstration.

The key audit conclusion was:

The production demo appears to be backed by saved offline experiment artefacts, not invented frontend records.

This was a significant academic integrity checkpoint.

10. Hardcoded Demo Selector

The audit found that the demo is pinned to selected saved experiment artefacts such as:

```
svd_top10_experiment  
seed99_robustness  
seed99_full_retry
```

This was classified as:

`HARDCODED_DEMO_SELECTOR`

This is not a data fabrication problem. It means the demo uses fixed saved artefact runs rather than allowing arbitrary dynamic experiment selection.

This was accepted as suitable for a dissertation artefact because it supports reproducible demonstration. However, it needed to be documented in README so that readers understand the page is displaying selected saved formal artefacts, not dynamically rerunning experiments.

11. High-Risk Field Audit: **description:keyword overlap**

A high-risk audit was performed for the field:

description:keyword overlap

This was important because the phrase sounded like an NLP or text-overlap calculation from item descriptions. Codex found:

- no exact string in frontend, backend, schema, API types, or saved artefacts;
- no backend calculation using **detail_desc**;
- no frontend derivation from description text;
- no saved keyword-overlap field;
- no matched field shaped like description keyword overlap.

Codex classified this as unsupported if it appeared in the UI.

A later screenshot of the explainability section confirmed that the current UI did not show **description:keyword overlap**. The displayed evidence chips were instead real metadata fields such as:

- product_group_name;
- product_type_name;
- colour_group_name;
- graphical_appearance_name;
- garment_group_name.

Conclusion:

description:keyword overlap is not currently visible and should not be used in the UI or dissertation.

No code modification was needed for this field.

12. High-Risk UI Field: **No hard constraints applied**

The 3-Agent Final Recommendation cards previously displayed:

CONSTRAINT STATUS

No hard constraints applied

This was identified as risky because the artefact does not implement a real hard-constraint recommendation layer.

The decision was made to remove this misleading field and replace it with a real H&M metadata field:

TYPE

This produced a safer and more consistent 2x2 metadata grid:

TYPE GROUP
COLOUR APPEARANCE

This was preferred over keeping a constraint-related label because the project’s contribution is not about constraint-based recommendation.

13. Audit of Four Metadata Fields

After replacing **Constraint Status** with **TYPE**, a read-only audit verified the source of the four displayed metadata fields.

The audit confirmed:

UI Field	Source	Classification
TYPE	product_type_name	real H&M metadata
GROUP	product_group_name	real H&M metadata
COLOUR	colour_group_name	real H&M metadata
APPEARANCE	graphical_appearance_name	real H&M metadata

It also confirmed:

Field	Source	Classification
MATCH SCORE	saved recommendation artifact score	real saved artefact
MATCHES count	derived from matched_evidence.length	frontend-derived from real array
score explanation	saved recommendation_reason explanation_text	/ real saved artefact
evidence chips	saved matched_evidence array	real saved artefact

This confirmed that the revised card layout is academically safe.

14. README / Artefact Provenance Documentation

A read-only documentation audit identified [README.md](#) as the best place for a short artefact provenance note.

A provenance note was added or prepared for README to explain that:

- [/artifact-demo](#) is a runnable offline dissertation artefact demo;
- it has three method views: SVD, 3-Agent, and Hybrid;
- it is pinned to selected saved formal experiment artefacts;
- it does not rerun experiments;
- it does not generate live recommendations;
- H&M metadata fields are real article metadata fields;
- recommendation scores, match scores, match counts, rank shifts, and explanation text are project-generated saved artefact outputs or derived presentation fields;
- static UI labels and fallback text are presentation-layer copy;
- the artefact does not claim CTR, CVR, conversion, live customer engagement, add-to-cart, dwell time, or live feedback adaptation.

The README was also reviewed for older framing. Older phrases such as “Agentic AI Recommendation Demo” and “LLM integration for user-intent interpretation and recommendation explanations” were identified as potentially outdated or risky if they made [/artifact-demo](#) sound like a live LLM-driven recommender. The recommended safer framing was:

H&M Hybrid SVD + 3-Agent Recommendation Artefact

and:

[/artifact-demo](#) is read-only and uses saved offline experiment and explainability artefacts.

15. Validation and Commit

After the relevant UI and documentation stabilisation work, final validation was completed.

The validation stage was intended to confirm:

- TypeScript type safety;
- frontend build stability;
- no backend changes unless explicitly needed;
- no experiment reruns;
- no saved artefact modifications;
- no unintended changes to recommendation logic or metrics.

Relevant files were committed after validation. This moved the project into packaging mode.

16. Current Status by 20 June

By 20 June, the project status was:

Core artefact: built and supervisor-recognised
/artifact-demo: working
SVD / 3-Agent / Hybrid tabs: present
Hybrid explainability layer: present
Source audits: completed for key high-risk fields
Constraint Status issue: fixed
H&M metadata field mapping: verified
description:keyword overlap: checked and not used
README provenance: added/reviewed
Validation: completed
Relevant files: committed
Header control row redesign: intentionally skipped
Current phase: ZIP packaging and supervisor submission

17. Remaining Work

The remaining work is not further system development. The remaining work is packaging and submission preparation.

Next tasks:

1. Confirm final git status and latest commit.
 2. Create a clean ZIP package.
 3. Include code, data-processing scripts, saved processed artefacts, explainability outputs, and references.
 4. Exclude `.git`, `node_modules`, `.venv`, `.next`, logs, `.tmp`, old reference folders, API keys, and unrelated prompt files.
 5. Ensure README explains artefact provenance and limitations.
 6. Send ZIP to supervisor.
 7. Prepare one results table for the next call showing SVD, 3-Agent, and Hybrid side by side on the same evaluation set with HR@10, NDCG@10, ILD@10, and confidence intervals.
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18. Academic Integrity Position

The final artefact should be described as:

A controlled offline H&M recommendation artefact comparing SVD, a structured 3-Agent workflow, and a Hybrid SVD + 3-Agent reranker with source-grounded explainability.

It should not be described as:

- a live recommender
- a fully autonomous agent system
- a system that improves CTR or CVR
- a system that proves live customer engagement improvement
- a system that replaces SVD
- a system that improves diversity

The correct conclusion remains:

The Standalone 3-Agent method does not replace SVD. The Hybrid method is best understood as an explainability-oriented reranking and augmentation layer over SVD. It improves substantially over the Standalone 3-Agent method, remains competitive with SVD, and provides source-grounded explanation evidence, while also trading off diversity.

This is the academically honest version of the project story and should guide the final report, viva, and supervisor discussion.